Shijie **Liu**

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Education

Chongqing University — *Chongqing, China*
PH.D. STUDENT IN WIND ENGINEERING — *Sep. 2024 - Present*

Shijiazhuang Tiedao University — *Shijiazhuang, China*
M.ENG. IN WIND ENGINEERING — *Sep. 2021 - Jun. 2024*
- Thesis: Data-driven reduced-order modelling methods for unsteady aerodynamics and fluid flows of the flow around bluff bodies.

Hebei University of Science and Technology — *Shijiazhuang, China*
B.ENG. IN CIVIL ENGINEERING — *Sep. 2017 - Jun. 2021*

Research Interests

`Core research interests:` I focus on **computational fluid dynamics**, **bluff-body aerodynamics**, and **particle-laden flows**. I study vortex shedding, wake transition, turbulent transport, inertial-particle clustering, void formation, and deposition using high-fidelity simulation and Lagrangian particle tracking. I also develop **reduced-order and data-driven methods**, including dynamic mode decomposition and LSTM-based models, for flow-field reconstruction and unsteady aerodynamic prediction.

`Core research question:` How do coherent wake structures govern unsteady aerodynamic loads and the transport of inertial particles around bluff bodies?

Honors and Awards (Selected)

2023-2024 **Top Ten Academic Stars of Graduate Students**, Shijiazhuang Tiedao University. [Link]
2022–2023 **National Scholarship for Graduate Students**, Ministry of Education of China. [Link]
2021-2022 **Excellent Graduate Student**, Top 10% of students at STDU.
2023 **Excellent Paper Award**, 32nd National Conference on Structural Engineering. [Link]
2019-2020 **Third Prize**, Pinming Cup National BIM Competition.
2018-2019 **First-Class Academic Scholarship**, Top 5% of students at HEBUST.

Publications (*equal contribution; †corresponding author)

PREPRINTS

[P1] xx
Shijie Liu, xx.
Under review at Physics of Fluids, 2026. [Paper]

JOURNAL ARTICLES

[J5] Effects of angle of attack and feature-preserving reduced-order models for canonical bridge deck wakes
Shijie Liu, Yuexin Cao, Zejun Qin, Jian Zhao, Luming An, Peng Guo, Zhen Zhang, and Qingkuan Liu.
Applied Sciences, 15(23):12670, 2025. [Paper]

[J4] Reconstruction and short-term prediction of wind pressure field on cylindrical coal sheds under undisturbed conditions using dynamic mode decomposition
Qingkuan Liu, Jing Huo, Shijie Liu, Zejun Qin, Haohan Li, and Zhen Zhang.
Journal of Wind Engineering and Industrial Aerodynamics, 258:106017, 2025. [Paper]

[J3] LSTM-based aerodynamic modeling for unsteady flows around structures
Shijie Liu, Zhen Zhang, Xue Zhou, and Qingkuan Liu.
Wind and Structures, 38(2):147–160, 2024. [Paper]

[J2] Bridge unsteady aerodynamic force prediction based on long short-term memory networks
Qingkuan Liu, Shijie Liu, Zhen Zhang, Xue Zhou, and Hongmiao Jing.
China Journal of Highway and Transport, 36(8):56–65, 2023. [Paper]

[J1] Unsteady flow-field analysis of flow around a square cylinder based on dynamic mode decomposition
Shijie Liu, Zhen Zhang, and Qingkuan Liu.
Engineering Mechanics, 2023. [Paper]

Conference Proceedings

[C3] DMD-based unsteady flow-field analysis around a square cylinder
Shijie Liu, Zhen Zhang, and Qingkuan Liu.
32nd National Conference on Structural Engineering, 2023. (Best Paper Award) [Paper]

[C2] LSTM-based aerodynamic coefficient prediction for two-dimensional cylinder flow
Shijie Liu, Bo Hu, Zhen Zhang, and Qingkuan Liu. [Paper]

[C1] Multi-scale LSTM prediction of wind pressure on coal-shed surfaces
Jing Huo, Shijie Liu, Zhen Zhang, and Qingkuan Liu. [Paper]

Research Experience

Chongqing University *Chongqing, China*
Ph.D. Student in Wind Engineering, advised by Prof. Cruz Y. Li *Sep. 2024 - Present*
- **Research Topics:** particle-laden bluff-body wakes, computational fluid dynamics, and Lagrangian particle tracking.

Shijiazhuang Tiedao University *Shijiazhuang, China*
M.S. in Wind Engineering, advised by Prof. Qingkuan Liu and Prof. Zhen Zhang *Sep. 2021 – Jun. 2024*
- **Research Topics:** data-driven unsteady aerodynamics, dynamic mode decomposition, and reduced-order flow-field modeling.

Academic Services

Teaching Assistant	BEE20203 Fluid Mechanics for Built Environment and Energy Engineering, Chongqing University; 58 lecture hours and four practical classes, Fall 2026. [Link]
Reviewer	Physics of Fluids.
Conference Volunteer	22nd National Conference on Structural Wind Engineering and 8th National Wind Engineering Graduate Forum, Chongqing, 2025. [Link]
Conference Volunteer	2nd Academic Conference on Extreme Loads and Protection of Bridge Structures, 2023. [Link]

Skills

Programming	Python, C++, LaTeX
GPU & ML	CUDA, PyTorch, TensorFlow
CFD Solvers	OpenFOAM, Fluent, STAR-CCM+, IAMR; iFlowX (in-house GPU-accelerated CFD solver)
Pre/Post-processing	ICEM CFD, Tecplot, Paraview
Wind Tunnel Experiments	Boundary-layer wind-tunnel testing, pressure measurements, force measurements, vibration measurements
Languages	Chinese, English; CET-4: 532, CET-6: 513

Invited Talks

[T1] Python in Research: Applications and Practice
Academic Launch event, organized by the Chongqing University Graduate Student Union, October 28, 2026 (scheduled). [Link]

Press Coverage (Selected)

[M1] "Selected as One of Shijiazhuang Tiedao University's Top Ten Academic Stars"
by New Media Center, Shijiazhuang Tiedao University, April 26, 2024. [Link]

[M2] "Awarded the National Graduate Scholarship"
by Wind Tunnel Laboratory, Shijiazhuang Tiedao University, October 31, 2023. [Link]

[M3] "Awarded Hebei Provincial Graduate Innovation Funding"
by Wind Tunnel Laboratory, Shijiazhuang Tiedao University, January 3, 2023. [Link]